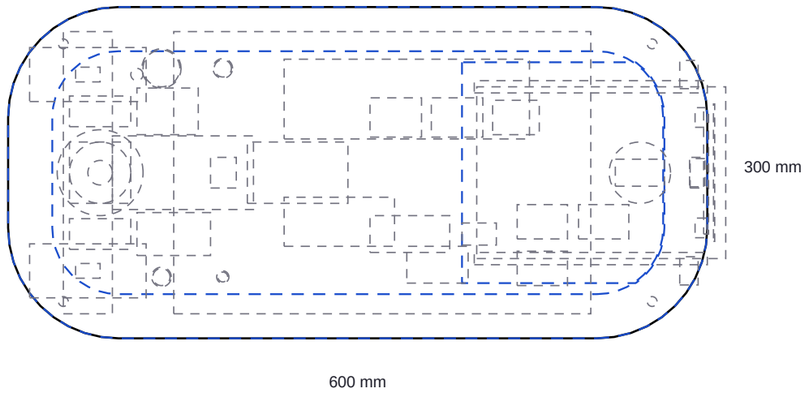
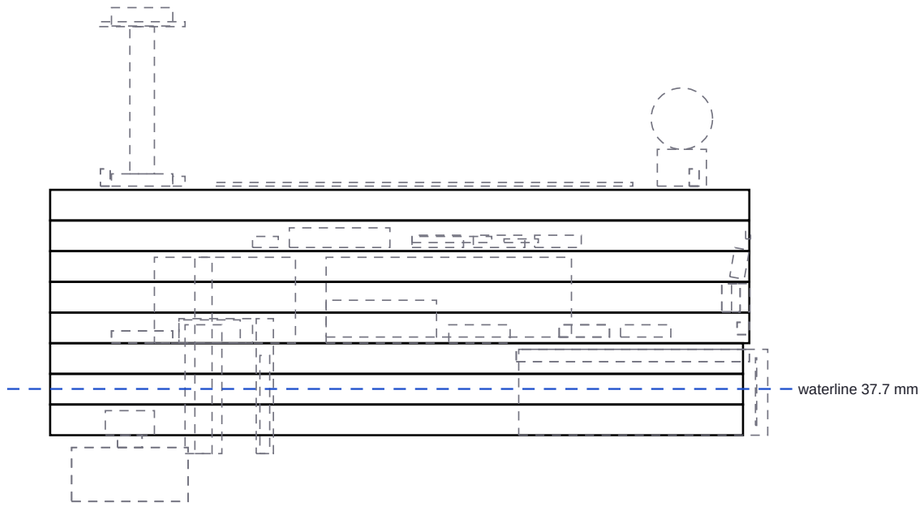
VARUNA AI Pod

600 x 300 mm thermocol boat | Your design, upgraded

PLAN (from above), scale 1:5



SIDE, scale 1:5

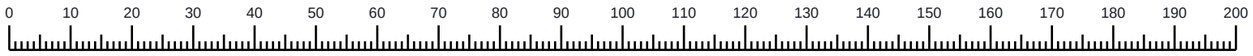


Size	600 x 300 mm, hull 200 mm tall, 348 mm to GPS top
Weight	4184 g empty, 4565 g with 381 g of plastic
Draft	37.7 mm empty, 40.9 mm full
Reserve buoyancy	10.5 kg before the deck goes under
Stability (GM)	104 mm: very stable
Trim	bow 2.6 mm higher than stern empty; slide the battery to level
Collection	1.9 L pull-out drawer
Camera	102 mm above water, tilted 10 deg down
Battery	Li-ion 3S6P 20 Ah (222 Wh), about 4.4 h
Thermocol	8 sheets 1000 x 500, or 3 sheets 2 x 3 ft (610 x 915)

How to use these templates

1. Print page 2 first at 100% / Actual size; the square must be 100 mm.
2. Big parts are HALF templates: the boat is the same left and right.
3. Join the two tiles, cut the paper out on the black line.
4. Draw a centreline on the thermocol. Lay the template's centreline on it, trace round it, flip the paper over, trace the other half.
5. Rings: trace the outside on 5 sheets. Cut the paper on the blue dashed line, trace it on layer 6; cut the red line, trace on 4 rings.
6. Mark holes by pushing a pencil through them. Small parts: cut and paste directly. Glue with Fevicol SH, never Fevikwik or spray paint.

cut line
cut out (window / pocket)
fold / glue line
placement guide, do not cut
skewer hole: lines up the layers



ruler in mm: the 0 to 200 marks must be exactly 200 mm apart

100 mm x 100 mm

How to print on A3 so every part comes out real size

1. Paper: A3, landscape. Printer: 'A3' selected as the paper size.
2. Scaling: 'Actual size' or '100%'. Turn OFF 'Fit to page', 'Shrink oversized pages' and 'Fit to printable area'. On a phone, choose 'Original size' / 'Do not scale'.
3. Print this page only. Measure the square with a steel ruler: both sides must be 100 mm, and the ruler marks must match your ruler. 1 mm off is fine; more is not.
4. If it is wrong, fix the printer setting and print this page again. Do not go on.
5. When it is right, print all the other pages with the SAME settings.
6. Big parts are split across pages (tiles A1, A2, B1...). Cut each tile on its border, lay the next tile over the grey overlap strip, line up the grey 50 mm grid and the measurement ruler, and tape it. Check the long measurement with a tape before cutting.
7. Glue-stick the paper on the thermocol, cut along the solid black line, cut out the red parts, and push a skewer through every + mark.

Line types:

- cut line
- cut out (window / pocket)
- fold / glue line
- placement guide, do not cut
- skewer hole: lines up the layers
- measurement (mm), do not cut

A1

570 overall length

135 (width 270 total)

wells: 125 and 175 from stern, 85 from centreline

left: turbidity 32 dia

right (flipped): pH 16 dia

left: TDS 16 dia

right (flipped): temperature 10 dia

layer 1 only:

beam slot

BASE LAYERS 1, 2, 3 - half template (cut 3, both halves)

0 50 100 150 200 250 300 350

CENTRELINE - flip the paper over here and trace the other half

overlap: tile A2 goes on top

570 overall length

R90 outside

R55 / R45 inside

135 (270 total)

RINGS - half template

red line: wall rings, layers 4, 5, 7 and lid rim 8 (cut 4)

blue dashed line: ledge ring, layer 6 (cut 1) - cut this line first

layer 5: also 3 ultrasonic holes at the bow (23 mm, see 3D)

50 100 150 200 250 300 350 400

45 ledge

35 wall

500 inside (wall)

CENTRELINE - flip here

A2

190 tunnel depth

75 (tunnel 150 total)

drawer tunnel (cut away)

400

450

500

35

Drawer tunnel roof (165 x 184 mm, 5 mm sunboard)

tank end

Ultrasonic board

Ultrasonic board

Ultrasonic board

Drawer back (134 x 67 mm, 3 mm sunboard)

cut out, cover with mesh

134 x 67

A1

570 overall length

135 (270 total)

nozzle 6 dia, 45 from stern

status LED 5 dia: LEFT SIDE ONLY

solar panel 340 x 230

LID TOP - half template, 3 mm sunboard

overlap: tile A2 goes on top

mast 21 dia, 75 from stern

CENTRELINE - flip here

panel cable 8 dia, 305 from stern

cable slot 30 x 25, both sides

SHELF - half template, 3 mm sunboard

495 long

99 (198 total)

Pixhawk on the centreline at 195 from stern; Pi Zero

CENTRELINE - flip here

A2

nozzle 6 dia, 525 from stern

camera cable 20 dia, 515 from stern

ern; Pi Zero right side

cut out, cover with mesh

203 x 134

Drawer front (entrance + handle) (134 x 85 mm, 3 mm sunboard)

handle

entrance: cut out

Camera wedge side (10 deg) (30 x 6 mm, 3 mm sunboard)

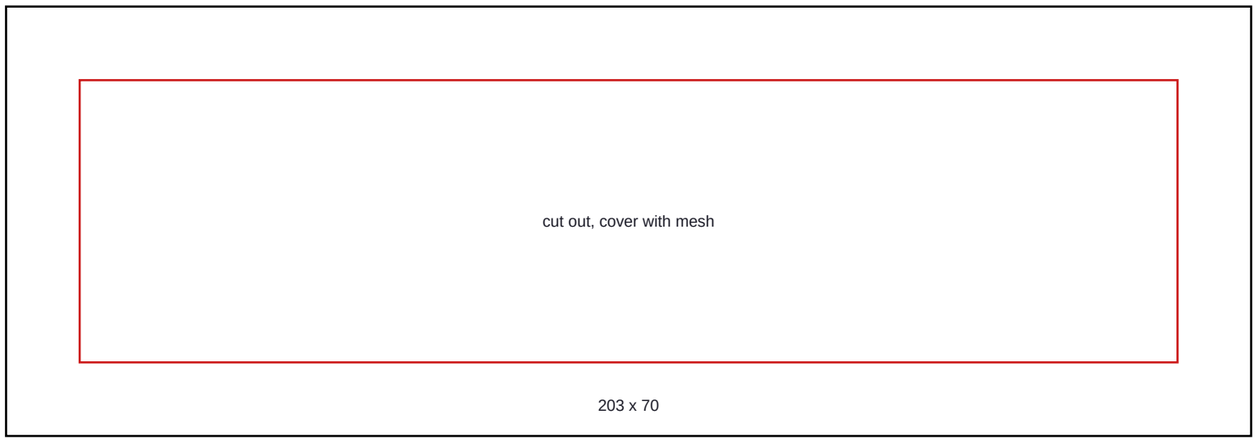
Camera wedge side (10 deg) (30 x 6 mm, 3 mm sunboard)

10 deg

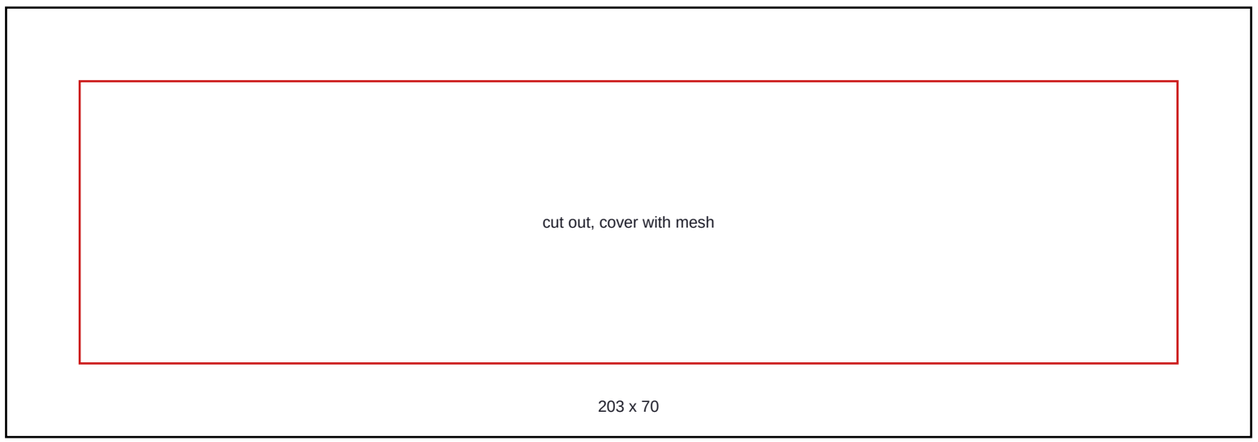
10 deg

<- must measure 50 mm

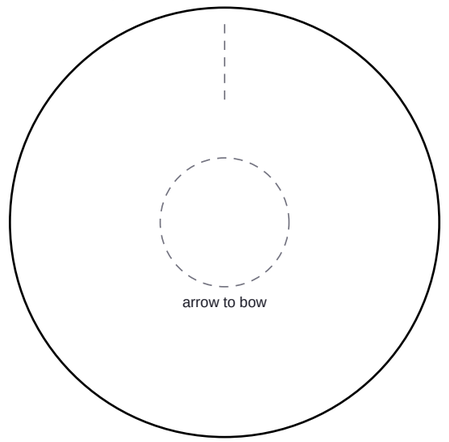
Drawer side (left)
3 mm sunboard | 203 x 70 mm



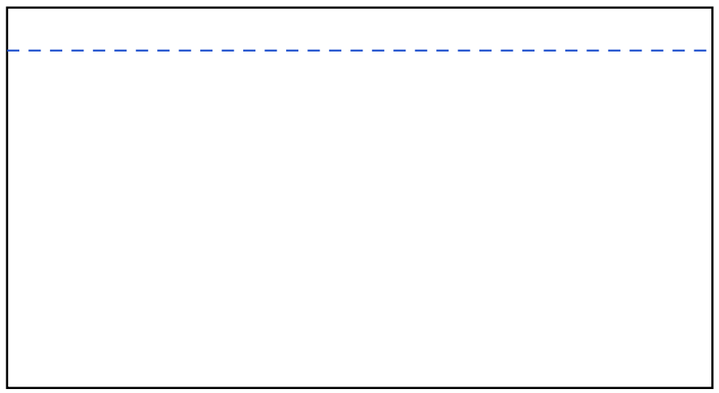
Drawer side (right)
3 mm sunboard | 203 x 70 mm



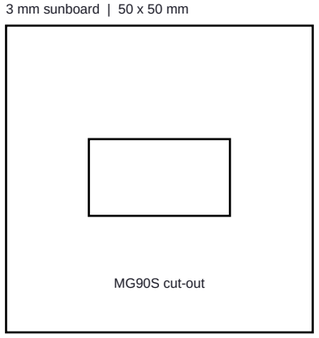
GPS disc
5 mm sunboard | 70 x 70 mm



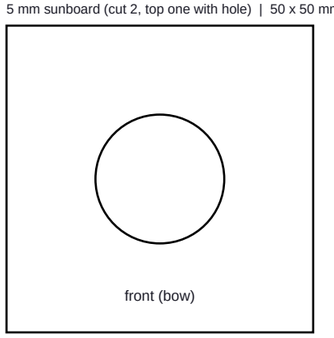
One-way flap
PET sheet | 115 x 62 mm



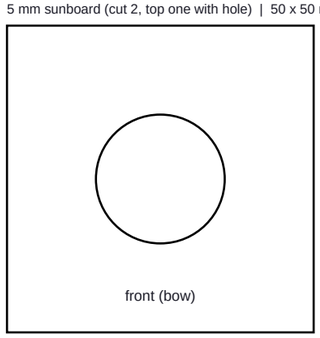
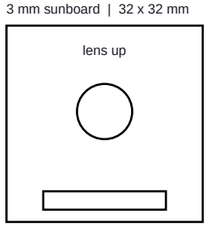
Servo plate
3 mm sunboard | 50 x 50 mm



Mast socket layer
5 mm sunboard (cut 2, top one with hole) | 50 x 50 mm



Mast socket layer
5 mm sunboard (cut 2, top one with hole) | 50 x 50 mm



Camera plate
3 mm sunboard | 32 x 32 mm